

# New vs Old Sample Pipeline Comparison: waveform-sim regeneration (run28)

PPG12 — isolated photon cross section in  $p+p$  at  $\sqrt{s} = 200$  GeV

*s***PHENIX** Internal

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## Abstract

The run28 MC trees were regenerated with the `g4waveform` branch's CEMC SiPM pixel-occupancy correction in `CaloWaveformSim`. This report compares the full nominal cross-section pipeline run on the **NEW** (regenerated) versus **OLD** (pre-regeneration) MC samples, walking a 10-item checklist with side-by-side overlays that reuse the existing plotting macros. The cross-section hardens: integral +1.7%, with a smooth  $p_T$ -tilt from  $-16\%$  at 8 GeV to  $+46\%$  at 34 GeV. The comparison localizes the cause to a single mechanism — the waveform correction shifts the simulated shower shapes, the BDT photon-ID score drops at high  $E_T$ , the ID and total efficiency fall above  $\sim 20$  GeV, and the efficiency-corrected spectrum hardens. ABCD purity and MBD-vertex efficiency are unchanged (new/old = 1.000). This is an internal MC-validation cross-check.

## 1 Method — sample-isolated comparison

To isolate the *sample* (tree) effect from any pipeline drift, the OLD trees (backed up at `/sphenix/tg/tg01/commiss`) were run through the *same current pipeline* under `var_type = bdt_nomold`. This `bdt_nomold` reference reproduces the May-15 result to  $< 0.5\%$  per bin, confirming the pipeline did not drift, so the differences below are genuinely from the waveform regeneration.

- **NEW** = `bdt_nom` (regenerated trees, current pipeline).
- **OLD** = `bdt_nomold` (pre-regen trees from backup, *same* current pipeline).
- Shower shapes: `ShowerShapeCheck` re-run on *both* tree sets with current code.

## 2 Headline result and causal chain

The cross-section **hardens**: integral **+1.7%**, with a smooth  $p_T$ -tilt from  **$-16\%$**  at 8 GeV to  **$+46\%$**  at 34 GeV. The comparison localizes the cause to a single mechanism:

SiPM-occupancy waveform correction  $\rightarrow$  **simulated shower shapes shift**  $\rightarrow$  **BDT photon-ID score drops at high  $E_T$**   $\rightarrow$  **ID (and total) efficiency lower above  $\sim 20$  GeV**  $\rightarrow$  efficiency-corrected cross-section **hardens**. ABCD **purity** and MBD-**vertex efficiency** are **unchanged** (new/old = 1.000), so they are *not* the driver. Energy scale ( $\sim 1.5\text{--}3\%$  lower) contributes via the unfolding response.

### 3 Cross-section: NEW / OLD (sample-isolated)

Table 1 gives the per-bin ratio of the efficiency-corrected, unfolded spectrum (`h_unfold_sub_result`, `Photon_final_bdt_nom.root` versus `Photon_final_bdt_nomold.root`).

| $E_T$ bin [GeV] | new/old | $E_T$ bin [GeV] | new/old              |
|-----------------|---------|-----------------|----------------------|
| 8–10            | 0.84    | 22–24           | 1.23                 |
| 10–12           | 1.01    | 24–26           | 1.31                 |
| 12–14           | 1.07    | 26–28           | 1.37                 |
| 14–16           | 1.10    | 28–32           | 1.37                 |
| 16–18           | 1.13    | 32–36           | 1.46                 |
| 18–20           | 1.18    | 36–45           | 2.16 (low stat)      |
| 20–22           | 1.19    | <b>integral</b> | <b>1.017 (+1.7%)</b> |

Table 1: Cross-section ratio NEW/OLD per  $E_T$  bin (sample-isolated). The hardening is smooth and monotonic above 10 GeV; the 36–45 GeV bin is low-statistics and should be ignored.

### 4 Headline figures

The five figures below trace the causal chain end-to-end, from the shower-shape driver (item 4) through the efficiency response (item 5) and energy scale (item 1b) to the unfolded spectrum (item 9) and the regenerated final cross-section (item 2).

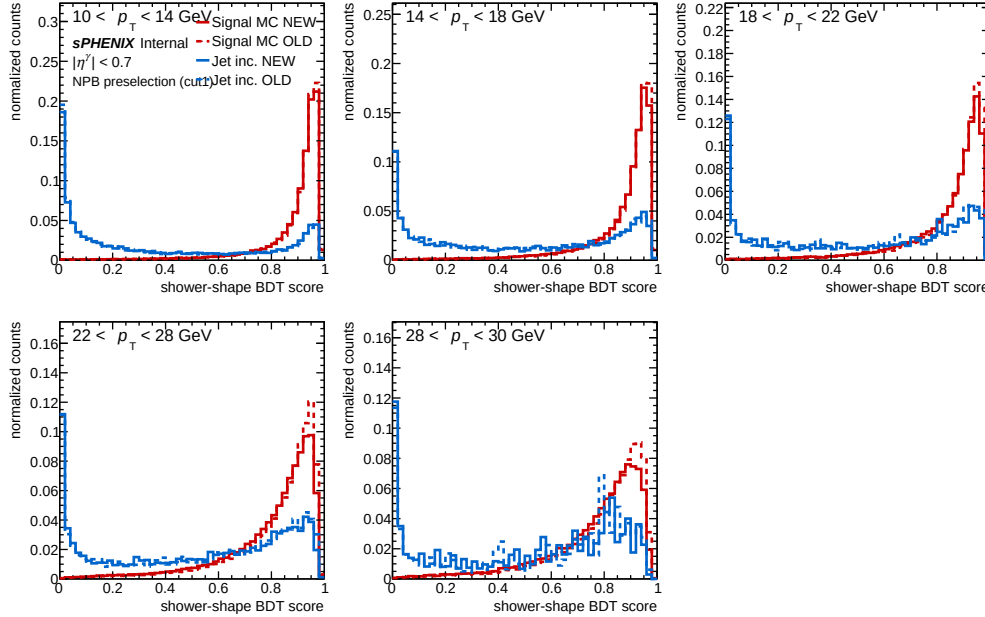


Figure 1: **[Item 4 — the driver.]** BDT photon-ID score, signal MC NEW versus OLD (1.5 mrad). The NEW signal BDT distribution shifts to lower values at high  $E_T$ , the direct cause of the ID-efficiency drop.

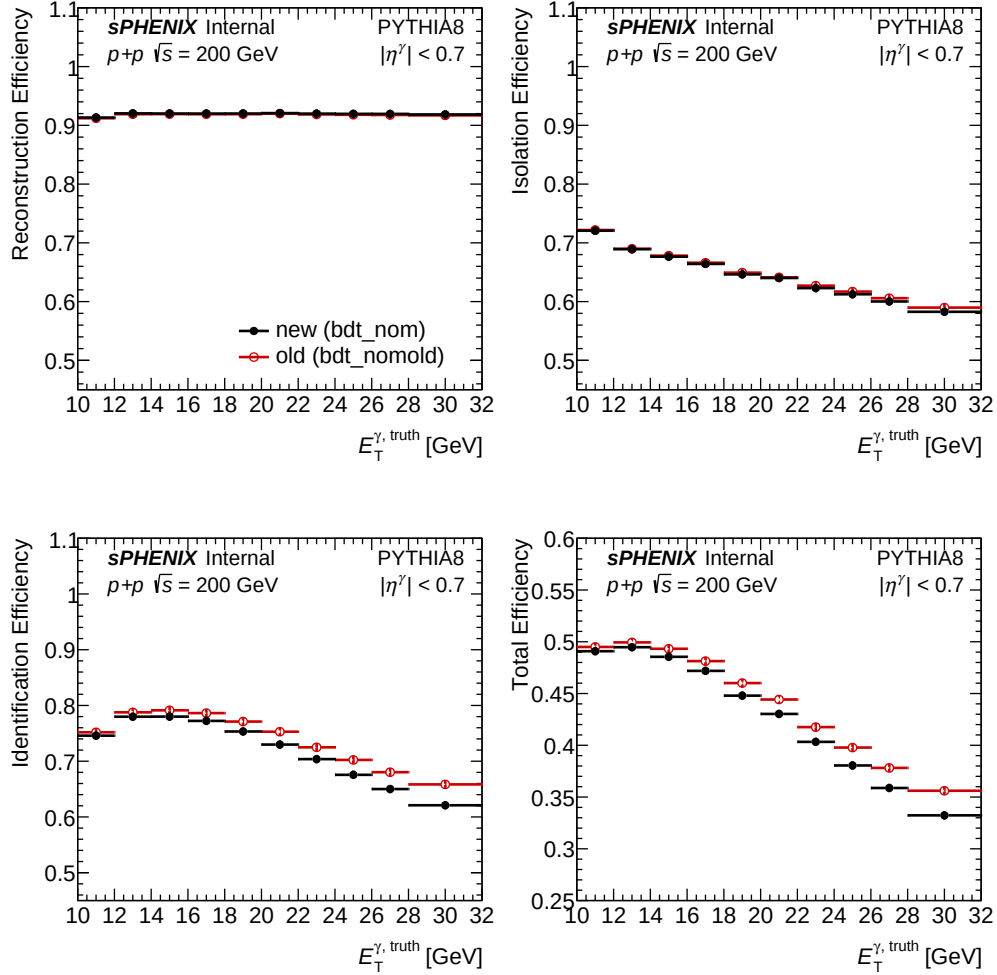


Figure 2: [Item 5.] Reconstruction, isolation, ID, and total efficiency NEW versus OLD. Reco and iso efficiency are unchanged; the ID and total efficiency are systematically lower above  $\sim 20$  GeV for NEW, propagating the BDT-score shift of Fig. 1 into the correction.

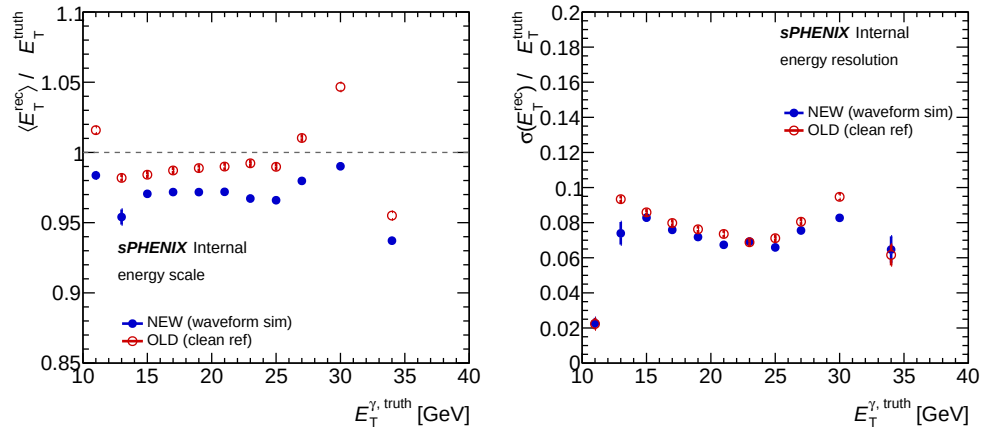


Figure 3: [Item 1b.] Energy scale and resolution NEW versus OLD. The NEW energy scale is  $\sim 1.5\text{--}3\%$  lower across 11–34 GeV and the resolution is slightly tighter; this feeds the hardening through the unfolding response.

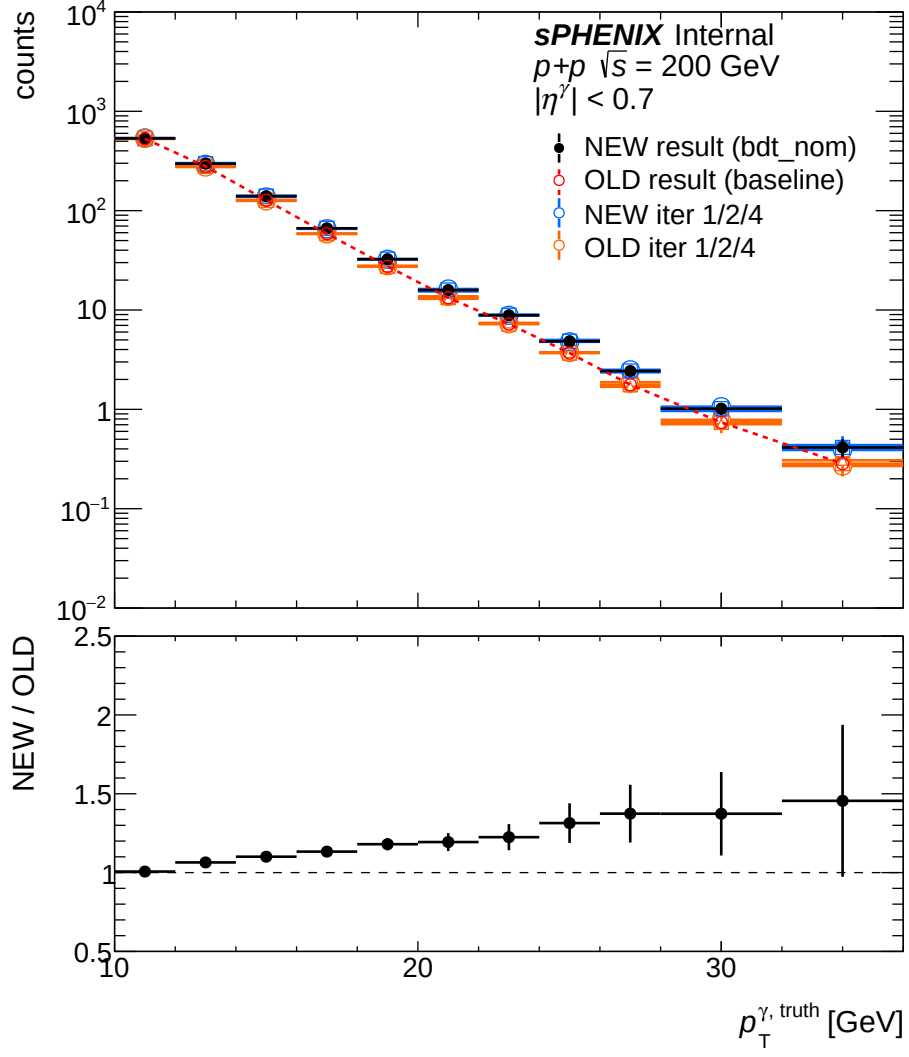


Figure 4: **[Item 9.]** Unfolded spectrum ratio NEW/OLD. The ratio is flat at low  $E_T$  and rises to  $\sim 1.4$ – $1.5$  at high  $E_T$ ; the iterations are stable, so the hardening is genuine and not an unfolding instability.

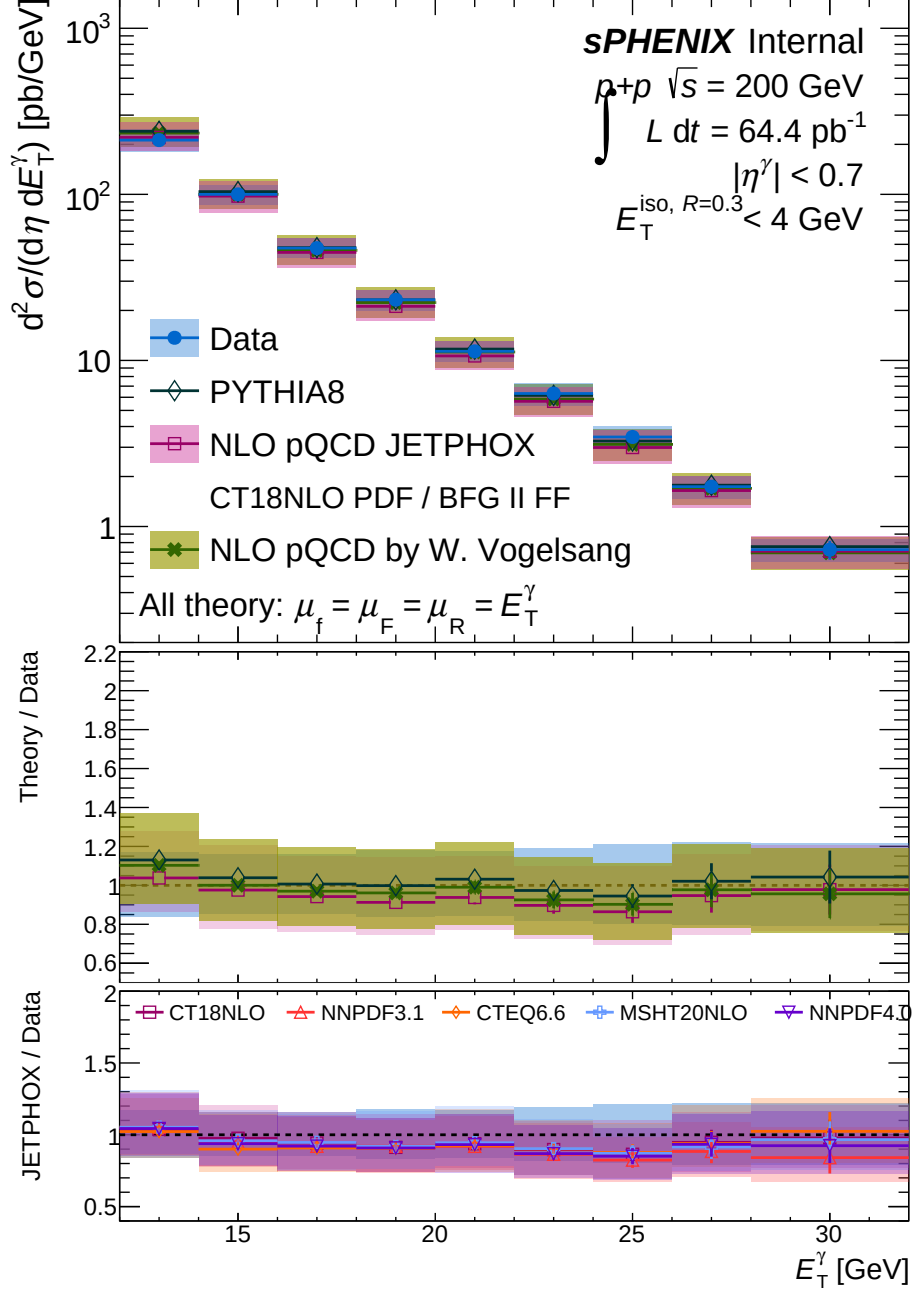


Figure 5: [Item 2.] Final cross section computed on the NEW (regenerated) trees. The spectrum hardens relative to the pre-regeneration result, and the Theory/Data ratio drops at high  $E_T$ .

## 5 Findings by checklist item

Table 2 summarizes all ten checklist items. Comparison figures live in `plotting/figures/compare/`; the headline figures above are referenced by their in-document label.

| #  | Item                       | Comparison figure                                                                                                         | Key finding (new vs old)                                                                                                                                    |
|----|----------------------------|---------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | Energy response matrix     | <code>response_compare.pdf</code>                                                                                         | NEW response carries more strength at high truth- $E_T$ (harder migration).                                                                                 |
| 1b | Energy scale & resolution  | <code>escale_eres_compare.pdf</code> , <code>escale_compare.pdf</code> , <code>eres_compare.pdf</code> (Fig. 3)           | NEW energy scale $\sim 1.5\text{--}3\%$ <b>lower</b> across 11–34 GeV; resolution slightly <b>tighter</b> .                                                 |
| 2  | Paper figures              | final x-sec<br><code>final_bdt_nom.pdf</code><br>(Fig. 5); eff/purity below                                               | Final spectrum hardens (Theory/Data ratio drops at high $E_T$ ).                                                                                            |
| 3  | Iso “fudging” factor       | n/a (config constant<br><code>mc_iso_scale=1.2</code> ,<br><code>mc_iso_shift=0.1</code> )                                | Constant, <b>unchanged</b> . Physics check = iso- $E_T$ shape, which falls out of #4.                                                                       |
| 4  | Shower shapes + BDT        | <code>showershape_{bdt,weta_cogx,mc_iso_scale,mc_iso_shift,mc_iso_rad,0rad}.pdf</code><br>(11 vars $\times p_T$ ; Fig. 1) | <b>NEW signal BDT score shifts to lower values at high <math>E_T</math></b> — the direct driver of the ID-eff drop.                                         |
| 5  | Efficiency                 | <code>eff_compare_{reco,iso,id,total,grid,sub}.pdf</code><br>(Fig. 2)                                                     | <b>reco, iso, id, and grid eff unchanged; ID &amp; total eff systematically lower above <math>\sim 20</math> GeV</b> for NEW; MBD eff unchanged.            |
| 6  | Signal leakage             | <code>leakage_compare.pdf</code>                                                                                          | leakage corrections $c_C$ , $c_D \sim 20\%$ larger at high $E_T$ for NEW; $c_B \sim 3\%$ larger.                                                            |
| 7  | Purity non-closure         | <code>mc_purity_correction_compare.pdf</code>                                                                             | ABCD purity itself <b>unchanged</b> (new/old = 1.000 per bin); non-closure correction $\sim$ stable.                                                        |
| 8  | Unfolding prior + reweight | <code>reweight_compare.pdf</code>                                                                                         | priors nearly identical; iter1/prior reweight ratio climbs <b>steeper</b> for NEW ( $\sim 1.55$ vs $\sim 1.15$ at 30 GeV).                                  |
| 9  | Unfolding response/closure | <code>unfold_iter_compare.pdf</code><br>(Fig. 4)                                                                          | unfolded NEW/OLD ratio flat at low $E_T$ rising to $\sim 1.4\text{--}1.5$ at high $E_T$ ; iterations stable (genuine hardening, not unfolding instability). |

Table 2: Findings for all ten checklist items. The five embedded headline figures are cross-referenced; the remaining comparison figures are named for lookup under `plotting/figures/compare/`.

## 6 Reviewer pass matrix

## 7 Files created / modified

- **Configs:** `efficiencytool/config_bdt_nomold{, _0rad, _1p5mrad}.yaml`, `efficiencytool/config_shower`, `efficiencytool/oneforall_tree_double_nomold_only.sub`, `showershape_di_jobs*_old.list`.

| Track                                     | Result                                                                                                                                                                 |
|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2b cosmetics<br>(plot-cosmetics-reviewer) | 11 figs reviewed: 9 PASS, 1 FAIL → <b>fixed</b><br>( <code>eff_compare_grid</code> layout), 2 minor WARN (non-blocking).<br>Shower-shape BDT figure spot-checked PASS. |
| 2c numerical re-derivation                | x-sec ratio, purity = 1.000, MBD = 1.000 re-extracted<br>directly from <code>Photon_final_bdt_nom.root</code> /<br><code>_bdt_nomold.root</code> .                     |
| 3.5 fix-validator                         | comparison macros re-ran end-to-end against the real<br><code>bdt_nomold</code> outputs (not the baseline stand-in).                                                   |

Table 3: Reviewer pass matrix for the comparison.

- **OLD-pipeline outputs (sample-isolated reference):** `efficiencytool/results/*_bdt_nomold*.root`, `MC_efficiencyshower_shape*_showershape*_old.root`.
- **Comparison macros (reuse existing plot code):** `plotting/plot_{response,reweight,unfold_iter,mc_purity_correction,escale,showershapes}_compare.C`.
- **Comparison figures:** `plotting/figures/compare/*.pdf` (all items above).
- **Baseline snapshot:** `efficiencytool/results/Photon_final_bdt_nom.preregen_baseline.root`.

## 8 Caveats

- The 36–45 GeV cross-section bin (+116%) is low-stat; ignore.
- The shower-shape OLD “background-only” hadd was skipped (the `bkgonly` job-list points only at NEW configs); the signal-combined and jet-inclusive overlays — which carry the comparison — are complete.
- Minor non-blocking cosmetics remain (legend crowding in `leakage_compare` panel-1 and the shower-shape header); fine for an internal cross-check, worth aligning with `strMC/strSigMC` before anything public.